Inference for Constant-Stress Partially Accelerated Life Test Model under Progressively Censored Unit Generalized Rayleigh Data

B.Tech Project Report

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DECLARATION BY THE STUDENT

I hereby certify that:

- The work presented in this thesis, entitled **"Inference for Constant-Stress Partially Accelerated Life Test Model under Progressively Censored Unit Generalized Rayleigh Data"** is my own and has been prepared under the guidance of **Dr. Yogesh Mani Tripathi**.
- This thesis has not been presented before to any institution or university for a degree or diploma.
- The work has been carried out in accordance with the instructions prescribed by the Institute for the preparation of theses.
- All the regulations regarding academic work of the institute have been followed.
- If any part of this thesis has been adopted from other work, a full reference for the same has been given in the bibliography. I have duly acknowledged the use of data, theoretical ideas, and other matters from the source.
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<u>CERTIFICATE</u>

*This is to certify that the work contained in this thesis entitled "**Inference for Constant-Stress Partially Accelerated Life Test Model under Progressively Censored Unit Generalized Rayleigh Data**" is a bonafide work of **RAVINDRA KUMAR NAYAK (Roll No. 2201MC30)**, carried out in the Department of Mathematics, Indian Institute of Technology Patna, under my supervision and guidance, and that it has not been submitted elsewhere for any degree.*

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RAVINDRA KUMAR NAYAK

Abstract

This thesis is focused on statistical inference about Constant-Stress Partially Accelerated Life Test (CSPALT) model with progressive Type-II censoring, considering the Unit Generalized Rayleigh (UGR) distribution as the lifetime model. The main goal is to make estimation about shape parameter α, scale parameter β, and acceleration factor λ with both classical and Bayesian approaches.

In classical inference, estimation by maximum likelihood estimation (MLE) and construction of asymptotic confidence intervals (ACIs) using the observed Fisher information matrix were made. A closed-form for the MLE of λ exists but the MLEs of α and β are obtained numerically. Bayesian estimation was conducted with two prior distributions: independent Gamma distributions and Gamma–Dirichlet prior that allows dependency between α and β. We obtained point estimations using both Squared Error Loss Function (SELF) and the LINEX loss function that is asymmetric. Since posterior distributions do not have closed-form solutions, the Bayesian estimations were obtained using the Metropolis-Hastings within Gibbs (MH-within-Gibbs) sampling technique. Bayesian credible intervals (BCIs) were constructed using samples of the posterior distribution obtained from simulation.

A Monte Carlo simulation study was carried out with two sets of parameters and seven progressively censored samples. The results are evaluated through average estimate (AE), mean squared errors (MSE), coverage probabilities (CP), and average interval length (AL). The results show that the Bayesian estimators yield smaller MSEs compared with the classical MLEs, especially under the Gamma-Dirichlet prior combined with the LINEX loss function. The BCI are narrower than ACIs while the coverage of the confidence interval are similar or even better in the Bayesian approach. Lastly, a practical data set that concerned on the breakdown time of insulating oil under both ordinary and acceleration tests was employed and the fit performance was analyzed with the UGR distribution.

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Chapter 1

Introduction

Reliability analysis concerns the nature and measurement of the failure behavior of engineering systems, electrical devices and industrial materials. One aspect of great practical importance is that complete failure data are seldom observed, because the test is terminated before all of the items have failed, or items are withdrawn during the test, or we simply lack the resources to continue observing all of the failures that occur in a test of an engineering system. All of these situations create censored data and it is important that censored observations be carefully incorporated into data analysis, so as to produce valid inferences.

Among various possible censoring schemes, progressively Type-II censoring which, in its simplest form, has n units on test at beginning and a prefixed number of survivors, say, removed at the occasion of each successive failure until the mth failure, has attracted considerable attention, since this mechanism achieves some balance between economic test costs and statistical efficiency, and it can facilitate the reuse of surviving test items in other experiments.

There is yet another issue of practical interest: Highly reliable product may fail after many years or decades, and so it is often desirable to accelerate failure processes in life testing under enhanced operating conditions, say elevated temperature, stress or voltage. Data collected at such stress level(s) will then be extrapolated to estimate the lifetime at normal conditions based on some appropriate models.

In absence of full knowledge of relationship between life and stress, or when it is technically or economically impossible or impractical to operate all units at same acceleration level, one has to be contented with some partially accelerated stress configuration. For example, a certain percentage of test units run under normal conditions and the rest are run under a fixed elevated stress. Then, these two groups are connected via an acceleration factor $\lambda > 1$, which is the ratio of the hazard rate under stress to that at normal conditions. This is the basic idea behind constant stress partially accelerated life test (CSPALT).

In this project, we are concerned about the Unit Generalized Rayleigh (UGR) distribution for lifetime modeling. UGR is a probability distribution defined on $(0, 1)$ and is generated through a logarithmic transformation on the Generalised Rayleigh distribution, whosecdf and pdf can be expressed in simple forms making likelihood computation simpler and whose hazard function is either initially increasing or bat-shaped. Due to its highly flexible nature in accommodating right-skewed failure time data, UGR becomes an attractive alternative choice compared to many other popular lifetime models.

Estimation of parameters (α, β, λ) are considered within both classical and Bayesian inference frameworks. For classical inference, maximum likelihood estimators (MLEs) were found using numerically calculated observed Fisher information matrix to construct asymptotic confidence intervals (ACIs). For Bayesian approach, the independence of parameters α, β, λ under Gamma priors is initially considered and then relationship between α, β is modelled using Gamma-Dirichlet prior structure. Point estimators under Squared Error Loss function and LINEX loss function were derived using MH-within-Gibbs algorithm for computing posterior distributions.

The efficiency of these estimators were explored through a comprehensive simulation study with and without the presence of real data from failure time of insulating oil under various stress level; Goodness-of-fit was addressed using Kolmogorov-Smirnov test.

Outline of the thesis. Chapter 2 describes UGR distribution, formulates the CSPALT model under progressive Type-II censoring. Chapter 3 describes classical approach; estimation is done using MLE and ACI is based on the observed information matrix. Chapter 4 describes Bayesian analysis with independent Gamma prior. Chapter 5 continues Bayesian analysis with Gamma-Dirichlet prior. Chapter 6 deals with the simulation study while chapter 7 presents real data analysis. Chapter 8 contains convergence diagnosis of Markov chains. Finally, chapter 9 gives the conclusion and future work.

Chapter 2

Model Description

2.1 Notations

We summarize the main notation in Table 2.1 to be used throughout the thesis.

Table 2.1: Summary of principal notation

Symbol	Description
T	Failure time (breakdown time) of a test unit
$F_j(t)$, $f_j(t)$	CDF and PDF under condition j ($j = 1$: normal, $j = 2$: accelerated)
$S_j(t)$, $H_j(t)$	Survival function and hazard rate function under condition j
α, β	Shape and scale parameters of the UGR distribution
λ	Acceleration factor ($\lambda > 1$)
n, m	Total units placed on test; total observed failures
n_j, m_j	Units and observed failures in group j, with $m_1 + m_2 = m$
x_{ji}	i-th ordered failure time from group j
$(S_{j1}, \ldots, S_{jm_j})$	Progressive Type-II censoring scheme for group j
$\hat{\alpha}, \hat{\beta}, \hat{\lambda}$	Maximum likelihood estimates
$\ell(\alpha, \beta, \lambda)$	Log-likelihood function
$I(\boldsymbol{\phi})$	Observed Fisher information matrix, $\boldsymbol{\phi} = (\alpha, \beta, \lambda)^T$

2.2 The Unit Generalized Rayleigh Distribution

Let X be the failure time of a test unit. The Unit Generalized Rayleigh (UGR) distribution is defined on $(0, 1)$ and its HRF, CDF, SF and PDF are defined under normal condition as below.

$$H_1(x \mid \alpha, \beta) = \frac{2\alpha\beta^2 \left(-\dfrac{\ln x}{x}\right) e^{-(\beta \ln x)^2}}{1 - e^{-(\beta \ln x)^2}}, \tag{2.1}$$

$$F_1(x \mid \alpha, \beta) = 1 - \left(1 - e^{-(\beta \ln x)^2}\right)^{\alpha}, \tag{2.2}$$

$$S_1(x \mid \alpha, \beta) = \left(1 - e^{-(\beta \ln x)^2}\right)^{\alpha}, \tag{2.3}$$

$$f_1(x \mid \alpha, \beta) = 2\alpha\beta^2 \left(-\frac{\ln x}{x}\right) e^{-(\beta \ln x)^2} \left(1 - e^{-(\beta \ln x)^2}\right)^{\alpha - 1}, \tag{2.4}$$

where $\alpha, \beta > 0$ and $0 < x < 1$. The standard relationships $S(x) = e^{-\int_0^x H(u)du}$, $F(x) = 1 - S(x)$ and $f(x) = H(x)S(x)$ are hold.

2.3 CSPALT Model Setup

The CSPALT with progressive Type-II censoring setup is defined as follows.

1. The failure times of the n test units are identically and independently distributed.
2. The n test units are split into two groups: n_1 running under normal condition (group 1) and $n_2 = n - n_1$ under an increased fixed condition (group 2).
3. The number of failures in each group j ($j = 1, 2$) is m_j, so that $m_1 + m_2 = m$, where m is the total number of failures observed. A fixed censoring scheme $\mathrm{CS}(S_{j1}, S_{j2}, \ldots, S_{jm_j})$ for the group j is prescribed where S_{ji} is the number of survivors removed at the i-th failure.
4. Under the accelerated condition, the hazard rate is scaled by the acceleration factor λ such that $H_2(x) = \lambda\, H_1(x)$ with $\lambda > 1$.

Using assumption 4 and equations (2.1)–(2.4) above, the hazard rate, survival function, CDF, and PDF under the accelerated condition are derived as:

$$H_2(x \mid \alpha, \beta, \lambda) = \frac{2\alpha\beta^2\lambda \left(-\dfrac{\ln x}{x}\right) e^{-(\beta \ln x)^2}}{1 - e^{-(\beta \ln x)^2}}, \tag{2.5}$$

$$S_2(x \mid \alpha, \beta, \lambda) = \left(1 - e^{-(\beta \ln x)^2}\right)^{\alpha\lambda}, \tag{2.6}$$

$$F_2(x \mid \alpha, \beta, \lambda) = 1 - \left(1 - e^{-(\beta \ln x)^2}\right)^{\alpha\lambda}, \tag{2.7}$$

$$f_2(x \mid \alpha, \beta, \lambda) = 2\alpha\beta^2\lambda \left(-\frac{\ln x}{x}\right) e^{-(\beta \ln x)^2} \left(1 - e^{-(\beta \ln x)^2}\right)^{\alpha\lambda - 1}. \tag{2.8}$$

2.4 Joint Likelihood under Progressive Type-II Censoring

Let x_{ji} $(i = 1, \ldots, m_j;\ j = 1, 2)$ denote the ordered observed failure times. The joint likelihood function for the progressively Type-II censored CSPALT data is

$$L(x) = \prod_{j=1}^{2} A_j \prod_{i=1}^{m_j} f_j(x_{ji}) \left[1 - F_j(x_{ji})\right]^{S_{ji}}, \tag{2.9}$$

where

$$A_j = n_j \left(n_j - 1 - S_{j1}\right)\left(n_j - 2 - S_{j2}\right) \cdots \left(n_j - m_j - \sum_{k=1}^{m_j - 1} S_{jk}\right).$$

Chapter 3

Classical Estimation

3.1 Maximum Likelihood Estimation

Equations (2.4), (2.3), (2.8), (2.6), and the joint likelihood (2.9) combined give a likelihood function (up to a positive constant)

$$\begin{aligned} L(\alpha,\beta,\lambda \mid x) \propto \prod_{i=1}^{m_1} & \left[2\alpha\beta^2\left(-\frac{\ln x_{1i}}{x_{1i}}\right) e^{-(\beta\ln x_{1i})^2}\left(1-e^{-(\beta\ln x_{1i})^2}\right)^{\alpha(S_{1i}+1)-1}\right] \\ & \times \prod_{i=1}^{m_2}\left[2\alpha\beta^2\lambda\left(-\frac{\ln x_{2i}}{x_{2i}}\right) e^{-(\beta\ln x_{2i})^2}\left(1-e^{-(\beta\ln x_{2i})^2}\right)^{\alpha\lambda(S_{2i}+1)-1}\right]. \end{aligned} \tag{3.1}$$

Neglecting additive constants, the log-likelihood is

$$\begin{aligned} \ell(\alpha,\beta,\lambda) =& m\ln(2\alpha\beta^2) + m_2\ln\lambda + \sum_{i=1}^{m_1}\big[-\ln x_{1i} + \ln(-\ln x_{1i}) - (\beta\ln x_{1i})^2 \\ & + (\alpha+\alpha S_{1i}-1)\ln\left(1-e^{-(\beta\ln x_{1i})^2}\right)\big] + \sum_{i=1}^{m_2}\big[-\ln x_{2i} + \ln(-\ln x_{2i}) - (\beta\ln x_{2i})^2 \\ & + (\alpha\lambda+\alpha\lambda S_{2i}-1)\ln\left(1-e^{-(\beta\ln x_{2i})^2}\right)\big]. \end{aligned} \tag{3.2}$$

Score Equations

By setting the first-order partial derivatives of (3.2) to zero, we get the score equations:

With respect to α:

$$\frac{\partial\ell}{\partial\alpha} = \frac{m}{\alpha} + \sum_{i=1}^{m_1}(1+S_{1i})\ln\left(1-e^{-(\beta\ln x_{1i})^2}\right) + \sum_{i=1}^{m_2}\lambda(1+S_{2i})\ln\left(1-e^{-(\beta\ln x_{2i})^2}\right) = 0. \tag{3.3}$$

With respect to λ:

$$\frac{\partial \ell}{\partial \lambda} = \frac{m_2}{\lambda} + \sum_{i=1}^{m_2} \alpha(1+S_{2i}) \ln\Big(1 - e^{-(\beta \ln x_{2i})^2}\Big) = 0. \tag{3.4}$$

With respect to β:

$$\begin{aligned}\frac{\partial \ell}{\partial \beta} &= \frac{2m}{\beta} - 2\beta \sum_{i=1}^{m_1} (\ln x_{1i})^2 - 2\beta \sum_{i=1}^{m_2} (\ln x_{2i})^2 + \sum_{i=1}^{m_1} (\alpha + \alpha S_{1i} - 1) \frac{2\beta (\ln x_{1i})^2 e^{-(\beta \ln x_{1i})^2}}{1 - e^{-(\beta \ln x_{1i})^2}} \\ &\quad + \sum_{i=1}^{m_2} (\alpha\lambda + \alpha\lambda S_{2i} - 1) \frac{2\beta (\ln x_{2i})^2 e^{-(\beta \ln x_{2i})^2}}{1 - e^{-(\beta \ln x_{2i})^2}} = 0. \end{aligned} \tag{3.5}$$

Closed-Form MLEs

From equations (3.3) and (3.4), the MLEs of α and λ can be expressed in closed form:

$$\hat{\lambda} = \frac{-m_2}{\sum_{i=1}^{m_2} \hat{\alpha}(1+S_{2i}) \ln\Big(1 - e^{-(\hat{\beta} \ln x_{2i})^2}\Big)}, \tag{3.6}$$

$$\hat{\alpha} = \frac{-m}{\sum_{i=1}^{m_1} (1+S_{1i}) \ln\Big(1 - e^{-(\hat{\beta} \ln x_{1i})^2}\Big) + \hat{\lambda} \sum_{i=1}^{m_2} (1+S_{2i}) \ln\Big(1 - e^{-(\hat{\beta} \ln x_{2i})^2}\Big)}. \tag{3.7}$$

MLE for β does not have a closed-form and has to be determined numerically. In the present work, equations (3.3)–(3.5) are jointly estimated by alternatively using the above formulas and a 1D search over $\hat{\beta}$.

3.2 Fisher Information Matrix

To construct asymptotic confidence intervals, the observed Fisher information matrix is used. For the parameter vector $\boldsymbol{\phi} = (\alpha, \beta, \lambda)^T$, define

$$I(\boldsymbol{\phi}) = - \begin{pmatrix} \frac{\partial^2 \ell}{\partial \alpha^2} & \frac{\partial^2 \ell}{\partial \alpha\, \partial \beta} & \frac{\partial^2 \ell}{\partial \alpha\, \partial \lambda} \\ \frac{\partial^2 \ell}{\partial \beta\, \partial \alpha} & \frac{\partial^2 \ell}{\partial \beta^2} & \frac{\partial^2 \ell}{\partial \beta\, \partial \lambda} \\ \frac{\partial^2 \ell}{\partial \lambda\, \partial \alpha} & \frac{\partial^2 \ell}{\partial \lambda\, \partial \beta} & \frac{\partial^2 \ell}{\partial \lambda^2} \end{pmatrix}_{(\hat{\alpha}, \hat{\beta}, \hat{\lambda})}. \tag{3.8}$$

For convenience, let $A_{1i} = 1 - e^{-(\beta \ln x_{1i})^2}$ and $A_{2i} = 1 - e^{-(\beta \ln x_{2i})^2}$. The non-zero second-order partial derivatives are:

$$\frac{\partial^2 \ell}{\partial \alpha^2} = -\frac{m}{\alpha^2}, \qquad \frac{\partial^2 \ell}{\partial \lambda^2} = -\frac{m_2}{\lambda^2}, \tag{3.9}$$

$$\frac{\partial^2 \ell}{\partial \alpha\, \partial \lambda} = \sum_{i=1}^{m_2} (1 + S_{2i}) \ln A_{2i}, \tag{3.10}$$

$$\frac{\partial^2 \ell}{\partial \beta\, \partial \alpha} = \sum_{i=1}^{m_1} (1 + S_{1i})\, \frac{2\beta (\ln x_{1i})^2\, e^{-(\beta \ln x_{1i})^2}}{A_{1i}} + \sum_{i=1}^{m_2} \lambda (1 + S_{2i})\, \frac{2\beta (\ln x_{2i})^2\, e^{-(\beta \ln x_{2i})^2}}{A_{2i}}, \tag{3.11}$$

$$\frac{\partial^2 \ell}{\partial \beta\, \partial \lambda} = \sum_{i=1}^{m_2} \alpha (1 + S_{2i})\, \frac{2\beta (\ln x_{2i})^2\, e^{-(\beta \ln x_{2i})^2}}{A_{2i}}, \tag{3.12}$$

and $\partial^2 \ell / \partial \beta^2$ involves an additional correction term from differentiating the ratio in (3.5) with respect to β. The variance–covariance matrix is $I^{-1}(\hat{\boldsymbol{\phi}})$, with diagonal elements $\widehat{\mathrm{Var}}(\hat{\alpha})$, $\widehat{\mathrm{Var}}(\hat{\beta})$, and $\widehat{\mathrm{Var}}(\hat{\lambda})$.

3.3 Asymptotic Confidence Intervals

Under regularity conditions, the MLEs are asymptotically normal:

$$(\hat{\alpha}, \hat{\beta}, \hat{\lambda}) \xrightarrow{d} N_3\Big((\alpha, \beta, \lambda),\ I^{-1}(\hat{\alpha}, \hat{\beta}, \hat{\lambda})\Big).$$

The $100(1 - \tau)\%$ asymptotic confidence intervals for the parameters are therefore

$$\begin{aligned} &\hat{\alpha} \pm z_{\tau/2} \sqrt{\widehat{\mathrm{Var}}(\hat{\alpha})}, \\ &\hat{\beta} \pm z_{\tau/2} \sqrt{\widehat{\mathrm{Var}}(\hat{\beta})}, \\ &\hat{\lambda} \pm z_{\tau/2} \sqrt{\widehat{\mathrm{Var}}(\hat{\lambda})}, \end{aligned} \tag{3.13}$$

where $z_{\tau/2}$ is the upper $(\tau/2)$-quantile of the standard normal distribution. Throughout this thesis the nominal level is 95%, so $\tau = 0.05$ and $z_{0.025} = 1.96$.

Chapter 4

Bayesian Estimation

4.1 Bayesian Estimation with Gamma Priors

For Bayesian Estimation we combine the likelihood observed with a prior on the unknown parameters. For this chapter we assume a Gamma prior for α and β and a truncated Gamma prior for λ.

4.1.1 Prior Distributions

The prior densities of α, β and $\lambda$are

$$\pi_1(\alpha) = \frac{b_1^{a_1}}{\Gamma(a_1)} \alpha^{a_1-1} e^{-b_1\alpha}, \quad \alpha > 0, \tag{4.1}$$

$$\pi_2(\beta) = \frac{b_2^{a_2}}{\Gamma(a_2)} \beta^{a_2-1} e^{-b_2\beta}, \quad \beta > 0, \tag{4.2}$$

$$\pi_3(\lambda) = \frac{b_3^{a_3}}{\Gamma(a_3,\, b_3 c)} \lambda^{a_3-1} e^{-b_3\lambda} \mathbb{I}_{\lambda \geq c}, \tag{4.3}$$

where $a_k, b_k > 0$ are hyperparameters ($k = 1, 2, 3$), $c = 1$ is the truncation point for λ, and $\Gamma(a_3, b_3 c) = \int_{b_3 c}^{\infty} u^{a_3-1} e^{-u}\, du$ is the upper incomplete Gamma function.
Here, the three parameters are assumed to be a priori independent, so the joint prior is

$$\pi(\alpha, \beta, \lambda) = \pi_1(\alpha)\, \pi_2(\beta)\, \pi_3(\lambda) \propto \alpha^{a_1-1} \beta^{a_2-1} \lambda^{a_3-1} e^{-(b_1\alpha + b_2\beta + b_3\lambda)}\, \mathbb{I}_{\lambda \geq 1}. \tag{4.4}$$

4.1.2 Posterior Distribution

Multiplication of equation (3.1) and (4.4) using Bayes' theorem gives the joint posterior

$$\pi(\alpha, \beta, \lambda \mid x) \propto L(\alpha, \beta, \lambda \mid x)\, \pi(\alpha, \beta, \lambda). \tag{4.5}$$

The conditional posteriors of α and λ are identifiable as Gamma (and truncated Gamma) distribution:

$$\pi(\alpha \mid \beta, \lambda, x) \sim \text{Gamma}(m + a_1,\ b_1 + B + C)\,, \tag{4.6}$$

$$\pi(\lambda \mid \alpha, \beta, x) \sim \text{Gamma}(m_2 + a_3,\ A) \quad (\lambda > 1), \tag{4.7}$$

where

$$B = -\sum_{i=1}^{m_1}(1 + S_{1i}) \ln\Big(1 - e^{-(\beta \ln x_{1i})^2}\Big)\,,$$
$$C = -\sum_{i=1}^{m_2} \lambda(1 + S_{2i}) \ln\Big(1 - e^{-(\beta \ln x_{2i})^2}\Big)\,,$$
$$A = b_3 - \sum_{i=1}^{m_2} \alpha(1 + S_{2i}) \ln\Big(1 - e^{-(\beta \ln x_{2i})^2}\Big)\,.$$

The conditional posterior of β does not have a standard distribution form so after manipulating terms we get

$$\begin{aligned}\pi(\beta \mid \alpha, \lambda, x) \propto \beta^{2m+a_2-1} \exp\Big[-b_2\beta - \sum_{i=1}^{m_1}(\beta \ln x_{1i})^2 - \sum_{i=1}^{m_2}(\beta \ln x_{2i})^2\Big] \\ \times \prod_{i=1}^{m_1}\Big(1 - e^{-(\beta \ln x_{1i})^2}\Big)^{\alpha(S_{1i}+1)-1} \prod_{i=1}^{m_2}\Big(1 - e^{-(\beta \ln x_{2i})^2}\Big)^{\alpha\lambda(S_{2i}+1)-1}.\end{aligned} \tag{4.8}$$

Sample for (4.8) are generated with the Metropolis–Hastings step within the Gibbs Algorithm.

4.1.3 Loss Functions

Squared Error Loss Function (SELF)

The SELF is a symmetric loss defined as

$$L_1(\hat{\theta}, \theta) = (\hat{\theta} - \theta)^2.$$

Under Squared Error Loss, the Bayes estimation is posterior mean $\hat{\theta}_{\text{SE}} = E(\theta \mid x)$.

LINEX Loss Function

With non-linear (asymmetric) costs from over-and under-estimation, we utilize the LINEX (Linear Exponential) loss function:

$$L_2(\Delta) = e^{s\Delta} - s\Delta - 1, \quad s \neq 0, \tag{4.9}$$

where $\Delta = \hat{\theta} - \theta$. When $s > 0$, overestimation is penalised heavily; when $s < 0$, underestimation is more costly. As $s \to 0$, the LINEX loss converges to the SELF.
The corresponding Bayes estimator is

$$\hat{\theta}_{\text{LINEX}} = -\frac{1}{s} \ln\big[E\big(e^{-s\theta} \mid x\big)\big] . \tag{4.10}$$

4.1.4 MH-within-Gibbs Sampling Algorithm

Since we can't obtain closed-form of the joint posterior (4.5), we simulate the posterior samples by using Metropolis-Hastings within Gibbs (MH-within-Gibbs) sampler. The steps of the algorithm 1 are as follows-.

Algorithm 1: MH-within-Gibbs Sampler for Gamma Priors

Input: Data x; hyperparameters a_k, b_k; total iterations N; burn-in M; proposal variance V_β

Output: Posterior samples $\{(\alpha^{(t)}, \beta^{(t)}, \lambda^{(t)})\}_{t=M+1}^{N}$

Initialisation: Set $(\alpha^{(0)}, \beta^{(0)}, \lambda^{(0)})$ to the MLE values.

for $t = 1, 2, \ldots, N$ **do**

  Step 1 (Sample α): Compute $B^{(t)}$ and $C^{(t)}$ using current $\beta^{(t-1)}$ and $\lambda^{(t-1)}$. Draw $\alpha^{(t)} \sim \text{Gamma}(m + a_1,\ b_1 + B^{(t)} + C^{(t)})$.

  Step 2 (Sample λ): Compute $A^{(t)}$ using $\alpha^{(t)}$ and $\beta^{(t-1)}$. Draw a candidate from $\text{Gamma}(m_2 + a_3,\ A^{(t)})$ and accept if it exceeds 1.

  Step 3 (Metropolis step for β): Propose $\beta^* \sim N(\beta^{(t-1)},\ V_\beta)$. Compute the acceptance probability

$$p = \min\left(1,\ \frac{\pi(\beta^* \mid \alpha^{(t)}, \lambda^{(t)}, x)}{\pi(\beta^{(t-1)} \mid \alpha^{(t)}, \lambda^{(t)}, x)}\right).$$

  Draw $u \sim U(0,1)$; set $\beta^{(t)} = \beta^*$ if $u \leq p$, otherwise $\beta^{(t)} = \beta^{(t-1)}$.

end

Post-processing: Discard the first M samples as burn-in.

4.1.5 Bayes Point Estimates

Let $\{(\alpha^{(t)}, \beta^{(t)}, \lambda^{(t)})\}_{t=M+1}^{N}$ denote the posterior samples after removing burn-in samples ($N - M$ samples in total). The Bayes estimates under SELF are the posterior means so:

$$\hat{\alpha}_{\text{SE}} = \frac{1}{N-M} \sum_{t=M+1}^{N} \alpha^{(t)}, \quad \hat{\beta}_{\text{SE}} = \frac{1}{N-M} \sum_{t=M+1}^{N} \beta^{(t)}, \quad \hat{\lambda}_{\text{SE}} = \frac{1}{N-M} \sum_{t=M+1}^{N} \lambda^{(t)}. \tag{4.11}$$

Under the LINEX loss function with parameter s:

$$\hat{\alpha}_{\text{LINEX}} = -\frac{1}{s} \ln \left[\frac{1}{N-M} \sum_{t=M+1}^{N} e^{-s\alpha^{(t)}} \right], \tag{4.12}$$

With corresponding terms for $\hat{\beta}_{\text{LINEX}}$ and $\hat{\lambda}_{\text{LINEX}}$.

4.1.6 Bayesian Credible Intervals

To construct a $100(1-\tau)\%$ Bayesian credible interval for a parameter θ, the preserved posterior samples are sorted in ascending order as $\theta_{(1)} \leq \theta_{(2)} \leq \cdots \leq \theta_{(N-M)}$. The interval is then

$$\left(\theta_{\left[(N-M)\tau/2 \right]},\ \theta_{\left[(N-M)(1-\tau/2) \right]} \right), \tag{4.13}$$

where $[\cdot]$ denotes the greatest integer function. For a 95% interval, $\tau = 0.05$, so the bounds are the 2.5th and 97.5th posterior percentiles.

4.2 Bayesian Estimation with Gamma–Dirichlet Priors

The GD prior places a more complex prior structure in place by modelling explicitly the dependence between the shape and scale parameters α and β. The truncated Gamma prior on λ stays the same as used in Chapter 4.

4.2.1 Gamma–Dirichlet Prior Structure

Let $\gamma = \alpha + \beta$. The GD prior is defined in 2 steps:

1. $\gamma \sim \text{Gamma}(a_0, b_0)$, with density

$$\pi_0(\gamma) = \frac{b_0^{a_0}}{\Gamma(a_0)} \gamma^{a_0-1} e^{-b_0\gamma}, \quad \gamma > 0.$$

2. Conditional on γ, $\alpha/\gamma \sim \text{Dirichlet}(a_1, a_2)$, with density

$$\pi_1\left(\frac{\alpha}{\gamma} \,\middle|\, \gamma \right) = \frac{\Gamma(a_1+a_2)}{\Gamma(a_1)\Gamma(a_2)} \left(\frac{\alpha}{\gamma} \right)^{a_1-1} \left(\frac{\beta}{\gamma} \right)^{a_2-1},$$

where $a_1, a_2 > 0$ are hyperparameters.

The marginal joint prior of (α, β) obtained by integrating out γ is

$$\pi(\alpha, \beta) = \frac{\Gamma(a_1 + a_2)}{\Gamma(a_1)\Gamma(a_2)} \, (\alpha + \beta)^{a_0 - a_1 - a_2} \, \alpha^{a_1 - 1} \, \beta^{a_2 - 1} \, e^{-b_0(\alpha+\beta)}. \tag{4.14}$$

When $a_0 = a_1 + a_2$, the term $(\alpha + \beta)^{a_0 - a_1 - a_2}$ equals one, and α and β become independent a priori, recovering the independent Gamma prior structure of Chapter 4. Otherwise, the two parameters are dependent, and the degree of dependence is governed by $a_0 - a_1 - a_2$.

The joint prior for all three parameters is $\pi(\alpha, \beta, \lambda) = \pi(\alpha, \beta) \, \pi_3(\lambda)$, where $\pi_3(\lambda)$ is the truncated Gamma prior of equation (4.3).

4.2.2 Posterior Distribution under GD Prior

Multiplying the likelihood (3.1) with the GD prior (4.14), then we get joint posterior as

$$\begin{aligned}\pi(\alpha, \beta, \lambda \mid x) \propto \alpha^{m+a_1-1} \, \beta^{2m+a_2-1} \, \lambda^{m_2+a_3-1} \, (\alpha+\beta)^{a_0-a_1-a_2} e^{-b_0(\alpha+\beta)-b_3\lambda} \\ \times \prod_{i=1}^{m_1} \left(-\frac{\ln x_{1i}}{x_{1i}} \right) e^{-(\beta \ln x_{1i})^2} \left(1 - e^{-(\beta \ln x_{1i})^2} \right)^{\alpha(S_{1i}+1)-1} \\ \times \prod_{i=1}^{m_2} \left(-\frac{\ln x_{2i}}{x_{2i}} \right) e^{-(\beta \ln x_{2i})^2} \left(1 - e^{-(\beta \ln x_{2i})^2} \right)^{\alpha\lambda(S_{2i}+1)-1}.\end{aligned} \tag{4.15}$$

The additional term $(\alpha + \beta)^{a_0 - a_1 - a_2}$, however, prevents the posterior from being a product of independent conditional distributions on α and β. For this reason, the posterior calculation using GD prior uses the standard MH-within-Gibbs algorithm (Algorithm 1) but with the updated posterior kernel (4.15) in the Metropolis acceptance ratio for both α and β steps.

4.2.3 Bayes Estimates and Credible Intervals

The Bayes point estimates for the parameters according to SELF and LINEX loss functions and Bayesian credible intervals under the GD prior can be found by using the formula of the parameters derived by postieror samples under GD prior in Sections 4.4 and 4.5 respectively (equations (4.11)–(4.13)).

Chapter 5

Simulation Study

5.1 Sample Generation Algorithm

Using Balakrishnan and Sandhu's (1995) procedure as adapted, Type-II progressively censored samples of the UGR distribution are produced under each stress condition as follows:

Algorithm 2: Generation of Type-II Progressively Censored Samples

Input: Sample size m; censoring scheme $(S_1, S_2, \dots, S_m)$; distribution functions F_1^{-1} and F_2^{-1}

Output: Progressively censored samples $\{X_i^{(1)}\}_{i=1}^m$ and $\{X_i^{(2)}\}_{i=1}^m$

Step 1 (Generate uniforms): Generate

$$W_1, W_2, \dots, W_m \sim \text{Uniform}(0, 1).$$

for $i = 1, 2, \dots, m$ **do**

Compute

$$V_i = W_i^{1/(i+S_m+S_{m-1}+\cdots+S_{m-i+1})}.$$

end

for $i = 1, 2, \dots, m$ **do**

Compute

$$U_i = 1 - V_m V_{m-1} \cdots V_{m-i+1}.$$

end

Step 4 (Generate censored samples): For $j = 1, 2$, set

$$X_i^{(j)} = F_j^{-1}(U_i), \qquad i = 1, 2, \dots, m,$$

where F_1^{-1} and F_2^{-1} denote the quantile functions under normal and accelerated conditions, respectively.

Output: The generated progressively Type-II censored samples $\{X_i^{(1)}\}$ and $\{X_i^{(2)}\}$.

Table 5.1: Progressive Type-II censoring schemes used in the simulation study

Scheme	(n, m)	Description
CS1	(n, m)	Standard Type-II: all $n - m$ removals at the last failure; $(0, 0, \ldots, 0, n - m)$
CS2	(n, m)	Early removal: all $n - m$ removals at the first failure; $(n - m, 0, 0, \ldots, 0)$
CS3	(n, m)	Spread removal: one unit removed at each of the first $n - m$ failures
CS4	(n, m)	Mixed progressive scheme
CS5	(n, m)	Early concentrated removal
CS6	(n, m)	Middle concentrated removal
CS7	(n, m)	General progressive scheme

We examine two choices for the parameter set:

$$\text{Setting 1:} \quad (\alpha, \beta, \lambda) = (1.6,\ 0.5,\ 2), \qquad \text{Setting 2:} \quad (\alpha, \beta, \lambda) = (2.5,\ 1.2,\ 3).$$

All results were obtained through 1,000 Monte Carlo replications. The performance is evaluated with:

- **Average Estimate (AE)**: $\mathrm{AE}(\hat{\theta}) = \dfrac{1}{N} \sum_{i=1}^{N} \hat{\theta}_i$.
- **Mean Squared Error (MSE)**: $\mathrm{MSE}(\hat{\theta}) = \dfrac{1}{N} \sum_{i=1}^{N} (\hat{\theta}_i - \theta)^2$.
- **Coverage Probability (CP)**: proportion of the N simulated intervals containing the true value.
- **Average Length (AL)**: $\mathrm{AL} = \dfrac{1}{N} \sum_{i=1}^{N} (U_i - L_i)$.

5.2 MLE Simulation Results

Tables 5.2 and 5.3 report the AE and MSE of the MLEs under Settings 1 and 2, respectively, across all seven censoring schemes.

Table 5.2: MLE: AE and MSE for ($\alpha = 1.6,\ \beta = 0.5,\ \lambda = 2$)

	$\hat{\alpha}$		$\hat{\beta}$		$\hat{\lambda}$	
Scheme	**AE**	**MSE**	**AE**	**MSE**	**AE**	**MSE**
CS1	1.797995	0.393293	0.513785	0.001593	2.089932	0.354617
CS2	1.759828	0.223539	0.510695	0.001393	2.130420	0.455987
CS3	1.852602	0.239326	0.515177	0.001194	2.049398	0.410953
CS4	1.728749	0.167980	0.509168	0.001591	2.094247	0.361747
CS5	1.783073	0.318199	0.507682	0.001344	2.085293	0.409691
CS6	1.826949	0.247279	0.510982	0.001247	2.017245	0.403579
CS7	1.783636	0.236105	0.509398	0.001333	2.061090	0.382028

Table 5.3: MLE: AE and MSE for ($\alpha = 2.5,\ \beta = 1.2,\ \lambda = 3$)

	$\hat{\alpha}$		$\hat{\beta}$		$\hat{\lambda}$	
Scheme	**AE**	**MSE**	**AE**	**MSE**	**AE**	**MSE**
CS1	2.899736	1.077981	1.230640	0.009018	3.272248	1.266280
CS2	2.608893	0.374109	1.211386	0.004610	3.265450	1.078949
CS3	2.620956	0.351525	1.209038	0.004574	3.164425	0.690418
CS4	2.718908	0.529651	1.210230	0.004683	3.211190	0.880780
CS5	2.702161	0.443070	1.214290	0.004018	3.210906	0.735931
CS6	2.771593	0.745097	1.225748	0.005519	3.264302	0.921750
CS7	2.797645	0.550304	1.219375	0.006200	3.065109	1.005429

5.3 ACI Results

Tables 5.4 and 5.5 report the coverage probability and average length of the 95% asymptotic confidence intervals.

Table 5.4: ACI: CP and AL for ($\alpha = 1.6,\ \beta = 0.5,\ \lambda = 2$)

	$\hat{\alpha}$		$\hat{\beta}$		$\hat{\lambda}$	
Scheme	**CP**	**AL**	**CP**	**AL**	**CP**	**AL**
CS1	0.94	1.754376	0.93	0.141236	0.96	2.198479
CS2	0.94	1.541216	0.93	0.132995	0.95	2.237047
CS3	0.96	1.644353	0.95	0.128552	0.91	2.131064
CS4	0.97	1.540427	0.93	0.128276	0.92	2.160920
CS5	0.92	1.563729	0.90	0.128527	0.94	2.158804
CS6	0.96	1.646502	0.95	0.133771	0.88	2.081651
CS7	0.93	1.593978	0.91	0.132826	0.96	2.158089

Table 5.5: ACI: CP and AL for ($\alpha = 2.5,\ \beta = 1.2,\ \lambda = 3$)

	$\hat{\alpha}$		$\hat{\beta}$		$\hat{\lambda}$	
Scheme	**CP**	**AL**	**CP**	**AL**	**CP**	**AL**
CS1	0.94	3.097494	0.95	0.314758	0.98	3.724981
CS2	0.97	2.395687	0.97	0.291123	0.95	3.603112
CS3	0.96	2.422412	0.95	0.278345	0.95	3.463994
CS4	0.97	2.569185	0.95	0.276700	0.97	3.528264
CS5	0.98	2.497040	0.97	0.281364	0.93	3.515994
CS6	0.96	2.653327	0.95	0.293879	0.95	3.605478
CS7	0.99	2.648853	0.96	0.290689	0.94	3.361953

5.4 Bayesian Simulation Results for Gamma Priors

5.4.1 Point Estimates under SELF

Table 5.6: Bayes–SELF (Gamma priors): AE and MSE for ($\alpha = 1.6,\ \beta = 0.5,\ \lambda = 2$)

	$\hat{\alpha}$		$\hat{\beta}$		$\hat{\lambda}$	
Scheme	**AE**	**MSE**	**AE**	**MSE**	**AE**	**MSE**
CS1	1.835313	0.303977	0.513847	0.001417	2.040460	0.257682
CS2	1.779643	0.194560	0.510307	0.001294	2.080982	0.277400
CS3	1.878224	0.219400	0.514927	0.001118	2.016855	0.329039
CS4	1.757959	0.172241	0.509962	0.001457	2.087062	0.288484
CS5	1.779022	0.251213	0.507477	0.001187	2.083633	0.270541
CS6	1.824769	0.196893	0.510767	0.001112	2.018249	0.213723
CS7	1.801270	0.208832	0.509358	0.001179	2.037003	0.251358

Table 5.7: Bayes–SELF (Gamma priors): AE and MSE for ($\alpha = 2.5,\ \beta = 1.2,\ \lambda = 3$)

	$\hat{\alpha}$		$\hat{\beta}$		$\hat{\lambda}$	
Scheme	**AE**	**MSE**	**AE**	**MSE**	**AE**	**MSE**
CS1	2.750996	0.604278	1.197805	0.005970	2.856485	0.562256
CS2	2.599396	0.336558	1.188863	0.004284	2.870135	0.385232
CS3	2.598786	0.281067	1.187158	0.004261	2.831184	0.442828
CS4	2.646405	0.326984	1.184046	0.004047	2.809841	0.429278
CS5	2.642141	0.298195	1.191598	0.003432	2.883798	0.375482
CS6	2.687961	0.493783	1.197205	0.003987	2.865849	0.425666
CS7	2.754350	0.415150	1.197477	0.004525	2.744169	0.543563

5.4.2 Point Estimates under LINEX Loss

Table 5.8: Bayes–LINEX ($s = 2$) (Gamma priors): AE and MSE for ($\alpha = 1.6,\ \beta = 0.5,\ \lambda = 2$)

	$\hat{\alpha}$		$\hat{\beta}$		$\hat{\lambda}$	
Scheme	**AE**	**MSE**	**AE**	**MSE**	**AE**	**MSE**
CS1	1.672233	0.153250	0.512675	0.001381	1.899592	0.209358
CS2	1.661819	0.125446	0.509251	0.001270	1.933628	0.221601
CS3	1.746738	0.125024	0.513949	0.001086	1.881756	0.292489
CS4	1.636400	0.110378	0.508980	0.001433	1.941363	0.224324
CS5	1.655610	0.156023	0.506473	0.001169	1.925901	0.217335
CS6	1.695914	0.118850	0.509698	0.001085	1.870324	0.178890
CS7	1.679744	0.132853	0.508308	0.001156	1.899817	0.219551

Table 5.9: Bayes–LINEX ($s = -2$) (Gamma priors): AE and MSE for ($\alpha = 1.6,\ \beta = 0.5,\ \lambda = 2$)

	$\hat{\alpha}$		$\hat{\beta}$		$\hat{\lambda}$	
Scheme	**AE**	**MSE**	**AE**	**MSE**	**AE**	**MSE**
CS1	2.042594	0.653902	0.515014	0.001456	2.188575	0.347398
CS2	1.920096	0.321225	0.511361	0.001320	2.244873	0.398908
CS3	2.043736	0.406899	0.515904	0.001153	2.178872	0.437438
CS4	1.915967	0.325824	0.510942	0.001483	2.247515	0.422509
CS5	1.927860	0.425823	0.508481	0.001208	2.263684	0.416575
CS6	1.984081	0.353293	0.511832	0.001142	2.177637	0.303868
CS7	1.957921	0.383560	0.510406	0.001204	2.187713	0.334367

Table 5.10: Bayes–LINEX ($s = 2$) (Gamma priors): AE and MSE for ($\alpha = 2.5,\ \beta = 1.2,\ \lambda = 3$)

	$\hat{\alpha}$		$\hat{\beta}$		$\hat{\lambda}$	
Scheme	**AE**	**MSE**	**AE**	**MSE**	**AE**	**MSE**
CS1	2.419626	0.331464	1.192936	0.005994	2.594667	0.549998
CS2	2.349454	0.234447	1.184377	0.004372	2.645259	0.424938
CS3	2.342157	0.194352	1.182931	0.004383	2.616469	0.503764
CS4	2.382140	0.202131	1.180029	0.004173	2.580463	0.471557
CS5	2.391987	0.180331	1.187365	0.003491	2.659825	0.406548
CS6	2.409891	0.323447	1.192649	0.004015	2.648627	0.462696
CS7	2.481163	0.231993	1.193016	0.004545	2.497647	0.600692

Table 5.11: Bayes–LINEX ($s = -2$) (Gamma priors): AE and MSE for ($\alpha = 2.5,\ \beta = 1.2,\ \lambda = 3$)

	$\hat{\alpha}$		$\hat{\beta}$		$\hat{\lambda}$	
Scheme	**AE**	**MSE**	**AE**	**MSE**	**AE**	**MSE**
CS1	3.161727	1.335807	1.202615	0.005993	3.116923	0.736586
CS2	2.919965	0.676055	1.193339	0.004240	3.097484	0.479921
CS3	2.928138	0.630548	1.191367	0.004177	3.061973	0.513515
CS4	2.980449	0.777551	1.188032	0.003955	3.034613	0.523101
CS5	2.957023	0.674924	1.195820	0.003410	3.110223	0.459378
CS6	3.050730	1.076420	1.201713	0.004006	3.104022	0.519161
CS7	3.085832	0.878114	1.201898	0.004546	3.003765	0.642781

5.4.3 Bayesian Credible Interval Results for Gamma Priors

Table 5.12: BCI (Gamma priors): CP and AL for ($\alpha = 1.6,\ \beta = 0.5,\ \lambda = 2$)

	$\hat{\alpha}$		$\hat{\beta}$		$\hat{\lambda}$	
Scheme	**CP**	**AL**	**CP**	**AL**	**CP**	**AL**
CS1	0.91	1.580898	0.93	0.132880	0.85	1.356828
CS2	0.90	1.362138	0.91	0.126917	0.85	1.420620
CS3	0.92	1.457875	0.95	0.122023	0.80	1.380537
CS4	0.93	1.402894	0.91	0.121804	0.84	1.386684
CS5	0.89	1.377387	0.89	0.123653	0.84	1.455167
CS6	0.91	1.435996	0.97	0.127400	0.84	1.403969
CS7	0.90	1.399115	0.91	0.125735	0.85	1.381710

Table 5.13: BCI (Gamma priors): CP and AL for ($\alpha = 2.5,\ \beta = 1.2,\ \lambda = 3$)

	$\hat{\alpha}$		$\hat{\beta}$		$\hat{\lambda}$	
Scheme	**CP**	**AL**	**CP**	**AL**	**CP**	**AL**
CS1	0.88	2.296692	0.94	0.267005	0.79	1.805109
CS2	0.93	2.021241	0.95	0.259612	0.74	1.689865
CS3	0.94	2.052342	0.92	0.250899	0.73	1.672406
CS4	0.94	2.064376	0.94	0.244817	0.76	1.708666
CS5	0.97	2.016422	0.96	0.251887	0.84	1.715113
CS6	0.91	2.131293	0.94	0.260715	0.77	1.714503
CS7	0.95	2.092814	0.95	0.257871	0.71	1.779744

5.5 Bayesian Simulation Results for GD Priors

5.5.1 Point Estimates under SELF

Table 5.14: Bayes–SELF (GD priors): AE and MSE for ($\alpha = 1.6,\ \beta = 0.5,\ \lambda = 2$)

	$\hat{\alpha}$		$\hat{\beta}$		$\hat{\lambda}$	
Scheme	**AE**	**MSE**	**AE**	**MSE**	**AE**	**MSE**
CS1	1.746919	0.237416	0.511024	0.001525	2.141511	0.350979
CS2	1.721087	0.160359	0.507009	0.001234	2.109170	0.273395
CS3	1.812515	0.175617	0.512511	0.001042	2.081336	0.342053
CS4	1.680951	0.117432	0.505280	0.001290	2.110819	0.298075
CS5	1.729259	0.224954	0.504877	0.001150	2.119988	0.315906
CS6	1.766143	0.144274	0.505683	0.001055	1.993676	0.243741
CS7	1.726788	0.168343	0.506524	0.001106	2.122818	0.334773

Table 5.15: Bayes–SELF (GD priors): AE and MSE for ($\alpha = 2.5,\ \beta = 1.2,\ \lambda = 3$)

	$\hat{\alpha}$		$\hat{\beta}$		$\hat{\lambda}$	
Scheme	**AE**	**MSE**	**AE**	**MSE**	**AE**	**MSE**
CS1	2.649363	0.580341	1.186665	0.005979	2.804716	0.424722
CS2	2.501957	0.230792	1.178074	0.003967	2.825053	0.336445
CS3	2.525334	0.284957	1.180949	0.004496	2.842409	0.382562
CS4	2.568014	0.316854	1.177532	0.003998	2.851050	0.364605
CS5	2.568642	0.278415	1.184488	0.003423	2.920184	0.440854
CS6	2.586008	0.363802	1.190078	0.004003	2.917606	0.431907
CS7	2.637008	0.316438	1.187412	0.005043	2.770607	0.555424

5.5.2 Point Estimates under LINEX Loss

Table 5.16: Bayes–LINEX ($s = 2$) (GD priors): AE and MSE for ($\alpha = 1.6,\ \beta = 0.5,\ \lambda = 2$)

	$\hat{\alpha}$		$\hat{\beta}$		$\hat{\lambda}$	
Scheme	**AE**	**MSE**	**AE**	**MSE**	**AE**	**MSE**
CS1	1.604300	0.145538	0.509891	0.001497	1.987213	0.253214
CS2	1.610017	0.108879	0.505952	0.001217	1.942137	0.190817
CS3	1.685155	0.108075	0.511509	0.001016	1.931062	0.274356
CS4	1.568714	0.086277	0.504293	0.001276	1.957104	0.236327
CS5	1.609167	0.147660	0.503873	0.001135	1.965978	0.239106
CS6	1.640454	0.084722	0.504575	0.001038	1.849975	0.198498
CS7	1.605680	0.108818	0.505448	0.001088	1.956948	0.266130

Table 5.17: Bayes–LINEX ($s = -2$) (GD priors): AE and MSE for ($\alpha = 1.6,\ \beta = 0.5,\ \lambda = 2$)

	$\hat{\alpha}$		$\hat{\beta}$		$\hat{\lambda}$	
Scheme	**AE**	**MSE**	**AE**	**MSE**	**AE**	**MSE**
CS1	1.932679	0.455822	0.512154	0.001556	2.299560	0.503686
CS2	1.854371	0.262805	0.508064	0.001252	2.284527	0.431690
CS3	1.972837	0.318337	0.513512	0.001070	2.247033	0.481286
CS4	1.816327	0.197101	0.506266	0.001305	2.280368	0.433817
CS5	1.880280	0.389796	0.505879	0.001166	2.292601	0.475534
CS6	1.925145	0.281000	0.506788	0.001076	2.154697	0.366543
CS7	1.880105	0.320641	0.507598	0.001126	2.299990	0.471323

Table 5.18: Bayes–LINEX ($s = 2$) (GD priors): AE and MSE for ($\alpha = 2.5,\ \beta = 1.2,\ \lambda = 3$)

	$\hat{\alpha}$		$\hat{\beta}$		$\hat{\lambda}$	
Scheme	**AE**	**MSE**	**AE**	**MSE**	**AE**	**MSE**
CS1	2.350452	0.380521	1.182099	0.006105	2.601513	0.502947
CS2	2.271137	0.204436	1.173545	0.004167	2.611310	0.423885
CS3	2.287252	0.231138	1.176774	0.004655	2.630793	0.439476
CS4	2.307552	0.228953	1.173215	0.004175	2.586000	0.447866
CS5	2.320002	0.206095	1.180181	0.003540	2.667005	0.440105
CS6	2.316240	0.261577	1.185233	0.004113	2.676416	0.490788
CS7	2.369981	0.210569	1.182785	0.005152	2.548437	0.590799

Table 5.19: Bayes–LINEX ($s = -2$) (GD priors): AE and MSE for ($\alpha = 2.5,\ \beta = 1.2,\ \lambda = 3$)

	$\hat{\alpha}$		$\hat{\beta}$		$\hat{\lambda}$	
Scheme	**AE**	**MSE**	**AE**	**MSE**	**AE**	**MSE**
CS1	3.050513	1.350730	1.191197	0.005897	3.021997	0.457050
CS2	2.781386	0.422924	1.182579	0.003810	3.056961	0.363910
CS3	2.825342	0.555024	1.185101	0.004374	3.063664	0.445870
CS4	2.906392	0.719418	1.181814	0.003861	3.110166	0.443906
CS5	2.895587	0.648535	1.188776	0.003346	3.173840	0.571429
CS6	2.939050	0.843647	1.194891	0.003947	3.180067	0.539023
CS7	2.986664	0.724980	1.192014	0.004973	3.008193	0.654806

5.6 Discussion of Simulation Results

There are several trends across all the simulation tables.

1. **MSEs are generally decreasing with sample size:** All the proposed methods exhibit this expected property, both for the classical and Bayesian estimates, indicating that the methods are consistently producing estimates.

2. **MSEs of Bayesian estimators are generally smaller than those of MLEs:** The MSEs of all the Bayesian estimates under Gamma and Gamma-Dirichlet priors are always smaller than their corresponding MLEs, except perhaps in some small specific cases for parameters and with various censoring schemes. Especially with the estimations of α and λ, the MSE for the MLE is considerably larger compared to the MLE of β.

3. **The best performing Bayesian estimator for λ is LINEX ($s = 2$):** For the parameter λ, the LINEX with $s = 2$ consistently yields the lowest MSE values for all the Bayesian estimators; the worst performing one for λ is the LINEX with $s = -2$, and the reason is that the loss function design tries to under-estimate to over-correct the estimation, which may cause over-estimation in a different way.

4. **Gamma-Dirichlet prior leads to a better estimation of α:** Since there is an assumed dependence between parameters α and β under the GD prior structure, the MSEs for parameter α under GD is more often than not smaller than those under the independent Gamma prior; while for parameter β and λ they are more or less the same with the parameters with independent Gamma prior since the prior form for λ is not changed.

5. **Length of intervals:** For almost all cases under different schemes and parameter settings, the ACIs tend to be wider than BCIs under both two different prior schemes, while coverage probabilities of BCIs are more or less the same or slightly lower than that of ACIs, which indicates the Bayesian credible interval might provide a better interval estimation in terms of coverage probability for this model.

6. **Censoring schemes are not sensitive to the proposed estimators:** Comparing the estimated results among the 7 different schemes for the same combination

of (n, m), the results vary only slightly. It appears that the performance of the proposed estimators is quite robust to the specific censoring process.

Chapter 6

Real Data Analysis

6.1 Dataset Description

In this chapter, we use the estimation procedures that have been developed on a real dataset, insulating oil breakdown time in CSPALT testing. This study has been conducted to explore the behavior of the proposed UGR model when applied to lifetime data under both normal and accelerated working stresses.

This data set contains two groups of measurements. The first set of values is the data from insulating oil samples tested under normal stresses, and the second set is the data from insulating oil samples tested under accelerated stresses. The breakdown-time data of these types are often used for lifetime data analysis, since insulating materials are expected to operate over a long time, and full failure information may take a long time to be observed.

The breakdown times for normal operating stresses are:

$$x_1 = \begin{cases} e^{-7.74},\ e^{-17.05},\ e^{-20.46},\ e^{-21.02},\ e^{-22.66},\ e^{-43.40},\ e^{-47.30}, \\ e^{-139.07},\ e^{-144.12},\ e^{-175.88},\ e^{-194.90}, \end{cases}$$

whereas the observations obtained under accelerated stress conditions are:

$$x_2 = \begin{cases} e^{-0.27},\ e^{-0.40},\ e^{-0.69},\ e^{-0.79},\ e^{-2.75},\ e^{-3.91},\ e^{-9.88},\ e^{-13.95}, \\ e^{-15.93},\ e^{-27.80},\ e^{-53.24},\ e^{-82.85},\ e^{-89.29},\ e^{-100.58},\ e^{-215.10}. \end{cases}$$

To account for censoring effects as presented in the simulation study, a Progressive Type-II censoring plan is used for both groups. The first $m_1 = 8$ failure time is used for inference for the normal-stress group, and the first $m_2 = 10$ failure time is used for inference for the accelerated-stress group. Hence, the two progressively censored samples are

$$x_1^{\text{obs}} = \Big\{ e^{-7.74},\ e^{-17.05},\ e^{-20.46},\ e^{-21.02},\ e^{-22.66},\ e^{-43.40},\ e^{-47.30},\ e^{-139.07},$$

and

$$x_2^{\text{obs}} = \begin{cases} e^{-0.27},\ e^{-0.40},\ e^{-0.69},\ e^{-0.79},\ e^{-2.75},\ e^{-3.91},\ e^{-9.88},\ e^{-13.95}, \\ e^{-15.93},\ e^{-27.80}. \end{cases}$$

The censored data above are then used to calculate both the maximum-likelihood and Bayesian estimates of the model parameters.

6.2 Goodness-of-Fit Analysis

Before parameter estimation, an evaluation is needed to determine whether the UGR distribution fits the data. For this purpose, the Kolmogorov–Smirnov goodness-of-fit test is applied individually to the normal and accelerated data sets.

The tests conducted are:

$$H_0 : \text{The data follows the UGR distribution}$$
$$H_1 : \text{The data does not follow the UGR distribution.}$$

The p-values obtained in both cases are $p_{\text{normal}} > 0.05, \quad p_{\text{accelerated}} > 0.05$. As can be observed, the p-values for both sets are greater than the critical value 0.05 used at a 5% level of significance, and as such, we do not reject the null hypothesis. Therefore, the UGR distribution is deemed appropriate to fit the insulating oil breakdown times in both instances.

6.3 Parameter Estimation

The values of the parameters α, β, and the acceleration factor λ are then estimated by the classical method (maximum likelihood estimate) and the Bayesian method based on the incrementally censored observations as presented above.

The maximum likelihood estimates of the unknown parameters are calculated based on the Newton–Raphson iterative technique as in Chapter 3. Bayesian estimates are calculated for the SELF and LINEX loss functions using the importance sampling technique derived in the Bayesian estimation chapter.

The point estimates and the interval estimates are presented in Table 6.1. The point and interval estimates contain: maximum likelihood estimates, Bayesian estimates for SELF and LINEX loss functions, and 95% Bayesian credible intervals, respectively.

Table 6.1: Parameter estimates and 95% interval estimates for the insulating oil breakdown-time data

Parameter	MLE	ACI Lower	ACI Upper	Bayes (SELF)	Bayes (LINEX)	BCI Lower	BCI Upper
α	0.221407	0.109842	0.362914	0.218077	0.214747	0.121951	0.347320
β	0.014043	0.007482	0.024118	0.015388	0.015375	0.008881	0.022717
λ	1.001000	0.998531	1.842615	1.177470	1.153980	1.004780	1.610367

6.4 Discussion of Results

From the estimation results given in Table 6.1, we can draw several conclusions.

Firstly, the maximum likelihood estimates of the parameters α and β are fairly close to their Bayesian estimates. This shows that the information contained in the observed data is sufficient to estimate the parameters reliably. In other words, the difference between classical and Bayesian estimates is small, and thus the prior does not overwhelm the data.

All three estimation procedures yielded a value less than 1 for the shape parameter α. This seems to reflect the observed data pattern, and thus the fitted distribution appears to capture well the skewness in breakdown times. The Bayesian credible interval for α is (0.121951, 0.347320). This looks like a reasonable range for the parameter within the context of the chosen Bayesian estimation approach.

The estimated values of the scale parameter β for the three estimation methods were again quite close. The credible interval is still fairly narrow, indicating that the parameter can be estimated with reasonable precision, even though the data were censored.

Special attention should be paid to the estimation of the acceleration factor λ. The MLE value $\hat{\lambda}_{\text{MLE}} \approx 1.001$ is very close to one, and this shows that the higher stress level may not have a significant effect on the intrinsic breakdown characteristics of insulating oil samples relative to the nominal operating stress level. Bayesian estimates with the SELF and LINEX loss functions are, however, 1.177470 and 1.153980, respectively. These results are expected in Bayesian estimation, which aims to avoid estimates at the edge of the parameter space by incorporating prior information into the estimation procedure. Credible interval for this parameter is (1.004780, 1.610367) and contains values considerably greater than one. Thus, some acceleration might be present.

Overall, the real data analysis supports the suggested estimators for CSPALT progressively censored data. The UGR distribution appears to provide a good fit for observed breakdown times, and the parameters estimated from classical and Bayesian methods are reasonably interpreted.

Chapter 7

Conclusion

In this thesis, statistical inference for the CSPALT model with progressive Type-II censoring has been studied under the UGR distribution. Both classical (MLE) and Bayesian procedures were proposed and examined through simulations and by analyzing real data.

On the classical side, we have an explicit form for the MLE of acceleration factor λ, and we needed to solve for the MLEs of α and β numerically by considering the system of nonlinear equations. Also we constructed confidence intervals based on the observed Fisher information matrix and they are reasonably close to nominal 95% confidence level in most situations.

On the Bayesian side, two different prior models were used; the prior was independent Gamma distributions on parameters α, β, λ and then the prior was Gamma–Dirichlet for parameters α and β, allowing dependence between α and β. We have not identified the conditional posterior distribution of β, hence, a MH-within-Gibbs sampler was used. We estimated the location of the posterior distribution under SELF and LINEX loss functions and the Bayesian credible interval was estimated from the posterior samples.

From the results of the simulation, we have clear trends; all Bayesian estimates have less MSE than their corresponding classical estimates, and this improvement is more significant on α and λ. Among the Bayesian estimation methods, LINEX estimator with $s = 2$ is the best for acceleration factor as over-estimation has been more heavily penalized. The joint Gamma–Dirichlet prior on α and β is better than independent prior on α and β in the sense of error for estimating α, because it accounts for the prior dependence between shape and scale parameters. The credible intervals obtained from Bayesian analysis had narrow intervals but good coverage, which seems to be better for interval estimation than classical intervals. For the application part on insulating oil failure times, the KS test shows that the distribution fits the data reasonably well for normal test as well as under accelerated test conditions. Also the estimate of acceleration factor for the data is approximately one indicating that this stress has limited effect on the failure mechanism of this data.

Areas for future scope. Several research opportunities arise from this work. First, the existing methods for choosing the censoring scheme can be optimized in many ways, such as minimizing the expected length of interval, or maximizing the amount of information collected during the censoring process. Second, step-stress PALT models where the stress changes at one or more times, could be studied under the same distribution for better representation of practical problems. Third, competing risk models and multi-component stress-strength problems for UGR distribution with progressive censoring should be studied.

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